Pneumatic Tool



Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Pneumatic tool

What it is used for

A pneumatic tool uses compressed air to perform tasks such as drilling, fastening, grinding, cutting, or impact work, depending on the tool type. Correct air-pressure settings, secure hose connections, suitable attachments, and safe handling help prevent injury and equipment damage.

Main parts

- Tool body houses the operating mechanism.
- Trigger or control lever starts and stops the tool.
- Air inlet connects the tool to the compressed-air supply.
- Air hose carries compressed air to the tool.
- Coupler and fittings connect the hose securely.
- Working attachment performs the specific task.
- Air regulator adjusts supply pressure where fitted.
- Exhaust port releases used air away from the operating mechanism.

Before you start

1. Only trained personnel should operate the pneumatic tool for the assigned task.
2. Read the tool manual and confirm its permitted applications and operating pressure.
3. Inspect the tool body, trigger, hose, couplers, fittings, and attachment for damage or leakage.
4. Check that the compressor, regulator, and hose are suitable for the tool's air demand and pressure rating.
5. Confirm that the attachment is compatible, correctly secured, and in good condition.
6. Wear task-appropriate eye protection, hearing protection, safety footwear, and other required PPE.
7. Secure the workpiece and clear the working area of people and loose objects.

How to use

1. Disconnect the air supply before fitting or changing the attachment.
2. Connect the hose using compatible couplers and verify that the connection is secure.
3. Set the regulator to the pressure specified by the tool manufacturer.
4. Check that the trigger is released before opening the air supply.
5. Open the air supply and check for leaks or unintended tool movement.
6. Hold the tool firmly using the designated grip and position the attachment correctly on the workpiece.
7. Operate the tool with controlled pressure and follow the recommended working technique.

8. Monitor tool performance, hose movement, vibration, and unusual noise during operation.
9. Release the trigger and shut off the air supply when the task is complete or an abnormal condition occurs.
10. Disconnect the air hose and release trapped pressure according to the tool instructions before changing attachments or cleaning.
11. Inspect the tool and workpiece, then clean and store the equipment safely.

⚠ Safety

- Never point a pneumatic tool or its exhaust toward yourself or another person.
- Disconnect the air supply and release trapped pressure before servicing or changing attachments.
- Never exceed the tool, hose, coupler, or accessory pressure rating.
- Do not use damaged hoses, loose fittings, or attachments with cracks or excessive wear.
- Keep hands and clothing away from rotating, cutting, or impact parts.
- Use guards and required PPE; do not disable safety controls or use compressed air to clean skin or clothing.

Care & storage

- Wipe the tool clean after use and remove debris from accessible areas.
- Inspect hoses, couplers, seals, triggers, and attachments regularly.
- Lubricate the tool only if required and only with the specified lubricant.
- Store the tool disconnected from the air supply in a clean, dry location.
- Arrange authorised repair if the tool leaks, loses power, or operates irregularly.

Common mistakes to avoid

- Operating at a pressure above the manufacturer's specified limit.
- Changing attachments while the air supply remains connected.
- Using damaged hoses or insecure couplers.
- Ignoring unusual vibration, air leakage, or a malfunctioning trigger.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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